

Lower bound program for $\lambda = (4, 3, 1^q)$: five-vector construction, A–B separation, and the position-of-letter contamination obstruction

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Abstract

The May-13 4-3-1n upper bound paper (2026-05-13-4-3-1n-upper-bound.tex) proved $\dim B_{j_0}^{(\lambda)} V_\lambda \leq 5$ structurally at $q \geq 5$ for $\lambda = (4, 3, 1^q)$, with computational verification of $\dim = 5$ at $q \in \{3, 4\}$. The matching *structural* lower bound $\dim \geq 5$ was deferred. This paper records partial progress toward that lower bound and identifies the precise technical obstruction.

What we establish. (i) An explicit five-vector construction: two vectors from Φ_A^λ lifting a basis of $B^{(\mu_A)}$ ($\mu_A = (3, 3, 1^{q-1})$, $\dim 2$), three vectors from Φ_B^λ lifting a basis of $B^{(\mu_B)}$ ($\mu_B = (4, 2, 1^{q-1})$, $\dim 3$). (ii) Structural *A vs B* separation via position-of- n : the exclusive cells $c_1 = (1, 4)$ for *A* and $c_2 = (2, 3)$ for *B* distinguish the two corner-pair contributions at the λ level, and the position-of- n projection commutes with the outer chain $\mathcal{O}_\lambda$. (iii) An analysis showing that the natural *within-X* position-of-letter separators (analogous to May-12 evening-3’s position-of- $(n-1)$ for $(4, 2, 1^*)$) do *not* propagate cleanly through the outer chain for either $X \in \{A, B\}$ of the $(4, 3, 1^q)$ shape.

The obstruction. For $(4, 2, 1^q)$ the outer chain has three factors $(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)$, and the separator position-of- $(n-1)$ is preserved by the first two factors, with only the last factor producing controlled motion via a single T_{n-2} -2-block. The cells reachable by this single move form a 2-*element set* that has clean exclusive elements. For $(4, 3, 1^q)$ the outer chain has *four* factors $(T_{n-5}+1) \cdots (T_{n-2}+1)$, the natural separator position-of- $(n-2)$ is moved by both $(T_{n-3}+1)$ and $(T_{n-2}+1)$, and the two-step motion enlarges the reachable cell set to include cells originally exclusive to the other branch. The contamination is not an artefact of the proof method but a structural fact: pos-of- $(n-2)$ at the λ level reaches the same set of cells starting from either inner basis vector.

Path forward. We identify the *tracer-pair* technique (this session’s memory note ‘tracer-pair-technique’) as the appropriate next tool: rather than separating F_A^1, F_A^2 via a position-of-letter projection, identify two specific λ -SYTs T_1^*, T_2^* and compute the seminormal coefficients $\langle v_{T_i^*}, F_A^j \rangle$ as products of Hoefsmit factors; exhibit a 2×2 tracer matrix with $\mathbb{Q}(q)^\times$ determinant. The chain of operations is finite and the computation is explicit; we sketch its structure but do not complete it in this paper.

Status. Partial. The A–B separation and the obstruction analysis are fully structural. The within-*X* linear independence reduces to a tracer-pair Hoefsmit computation that we leave for a follow-up paper.

1 Setup

Conventions follow the May-13 4-3-1n upper bound paper (2026-05-13-4-3-1n-upper-bound.tex, henceforth **May-13-UB**). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and the standard braid relations. V_λ is the seminormal cell module for

$\lambda \vdash n$ in asymmetric Murphy normalisation ($b' = 1$). $E_j^\pm$ denote the q - and (-1) -eigenspaces of T_j . For $1 \leq k \leq n$,

$$R'_k := (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1), \quad B_j^{(\lambda)} V_\lambda := R'_j R'_{j+1} \cdots R'_n V_\lambda.$$

Sharp index $j_0(\lambda) := \ell(\lambda) + 1$.

Throughout this paper. $\lambda := (4, 3, 1^q)$ with $q \geq 3$, $n := q+7$, $\ell := q+2$, $j_0 := q+3$. Removable corners

$$c_1 = (1, 4), \quad c_2 = (2, 3), \quad c_3 = c_* = (q+2, 1).$$

c_1, c_2 are length-preserving; c_3 is the column-1 bottom. Same-row pair at row 2: cells $(2, 2), (2, 3)$.

Inner partitions.

$$\begin{aligned} \mu_A &:= \lambda \setminus \{c_1, c_3\} = (3, 3, 1^{q-1}), & |\mu_A| &= q+5 \\ \mu_B &:= \lambda \setminus \{c_2, c_3\} = (4, 2, 1^{q-1}), & |\mu_B| &= q+5 \\ \mu_C &:= \lambda \setminus \{c_1, c_2\} = (3, 2, 1^q), & |\mu_C| &= q+5 \end{aligned}$$

Outer chain. $\mathcal{O}_\lambda := (T_{j_0-1}+1)(T_{j_0}+1)(T_{j_0+1}+1)(T_{n-2}+1) = (T_{q+2}+1)(T_{q+3}+1)(T_{q+4}+1)(T_{q+5}+1)$, four factors with indices $\ell, \ell+1, n-3, n-2$.

Recall: May-13-UB reduction.

$$B_{j_0}^{(\lambda)} V_\lambda = \mathcal{O}_\lambda \left[\Phi_A^\lambda (B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A}) + \Phi_B^\lambda (B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B}) \right]. \quad (1)$$

The μ_C -piece and same-row S -piece both vanish structurally for all $q \geq 3$ (May-13-UB Lemmas 3.5 and 3.6: j -formula leakage and the same-row-dies framework).

Inner basis inputs.

- $\dim B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} = 2$. Structural at $q \geq 5$ (SRD-331 + May-12 evening on the inner inner shape $\mu_1^A := \mu_A \setminus \{c_1(\mu_A), c_3(\mu_A)\} = (3, 2, 1^{q-2})$). Computational at $q \in \{3, 4\}$. Basis $\{u_A^1, u_A^2\}$ explicit (via lifting through the $r=1$ reduction; see Section 4.1).
- $\dim B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} = 3$. Structural at $q \geq 5$ (May-12 evening-3 + SRD-421). Computational at $q \in \{3, 4\}$ inputs. Basis $\{u_B^1, u_B^2, u_B^3\}$ explicit (May-12-Eve3 §5–7; see Section 4.2).

2 Five-vector construction

Definition 1. Define five vectors in V_λ :

$$\begin{aligned} F_A^i &:= \mathcal{O}_\lambda \Phi_A^\lambda(u_A^i), & i &\in \{1, 2\}, \\ F_B^j &:= \mathcal{O}_\lambda \Phi_B^\lambda(u_B^j), & j &\in \{1, 2, 3\}. \end{aligned}$$

By (1) together with the inner-basis spanning properties, $\{F_A^i\}_i \cup \{F_B^j\}_j$ spans $B_{j_0}^{(\lambda)} V_\lambda$. The conjecture is that these five vectors are *linearly independent*, which would give $\dim = 5$.

Theorem 2 (A–B decoupling reduces lower bound to two within-class projections). *Suppose*

(a) $\{\pi_n^{(c_1)} F_A^1, \pi_n^{(c_1)} F_A^2\}$ are linearly independent in $\pi_n^{(c_1)} V_\lambda$.

(b) $\{\pi_n^{(c_2)} F_B^1, \pi_n^{(c_2)} F_B^2, \pi_n^{(c_2)} F_B^3\}$ are linearly independent in $\pi_n^{(c_2)} V_\lambda$.

Then the five vectors $\{F_A^1, F_A^2, F_B^1, F_B^2, F_B^3\}$ are linearly independent in V_λ . In particular, $\dim B_{j_0}^{(\lambda)} V_\lambda \geq 5$.

The structural content of Theorem 2 is the reduction: the full 5-vector independence follows from *two independent* within-class projection independences. The A vs B coupling is broken by the position-of- n projection alone.

3 A vs B separation via position-of- n

3.1 Position-of- n is a $\mathcal{O}_\lambda$ -invariant

Lemma 3 (Position-of- n invariant under outer chain). *For every seminormal basis vector $v_T \in V_\lambda$ and every $1 \leq j \leq n-2$, $T_j v_T$ is a $\mathbb{Q}(q)$ -linear combination of seminormal basis vectors $v_{T'}$ with $\text{pos}_{T'}(n) = \text{pos}_T(n)$.*

Proof. T_j acts on the seminormal basis via the standard Hoefsmit formulae (same row $\Rightarrow q$, same column $\Rightarrow -1$, otherwise a 2-block on $\{v_T, v_{s_j T}\}$ with s_j swapping letters j and $j+1$). The position of letter n is unaffected by s_j since $j+1 \leq n-1$, so neither swap touches the letter n . $\square$ $\square$

Corollary 4 (Seminormal projection commutes with $\mathcal{O}_\lambda$). *For each cell c of λ , the projection $\pi_n^{(c)}$ onto seminormal vectors with $\text{pos}(n) = c$ commutes with each factor (T_j+1) , $1 \leq j \leq n-2$. Since the indices in $\mathcal{O}_\lambda$ are $\{q+2, q+3, q+4, q+5\} \subset \{1, \dots, n-2\}$ (as $n-2 = q+5$), $\pi_n^{(c)}$ commutes with $\mathcal{O}_\lambda$.*

3.2 Image of Φ_X^λ has restricted position-of- n support

Lemma 5 (Position-of- n support of $\Phi_X^\lambda(V_{\mu_X})$). *For each corner pair $X \in \{A, B, C\}$ of λ and every $v \in V_{\mu_X}$, every SYT T in the seminormal support of $\Phi_X^\lambda(v)$ has $\text{pos}_T(n) \in \{c_X^-, c_X^+\}$. In particular:*

- For $X = A$: $\text{pos}_T(n) \in \{c_1, c_3\} = \{(1, 4), (q+2, 1)\}$.
- For $X = B$: $\text{pos}_T(n) \in \{c_2, c_3\} = \{(2, 3), (q+2, 1)\}$.

Proof. By the construction of Φ_X^λ (May-13-UB Theorem 2.2, item 1): $\Phi_X^\lambda(v_{T'}^{(\mu_X)}) = v_{T_A^X} + \tau_X v_{T_B^X}$ where T_A^X, T_B^X are the two extensions of T' obtained by placing $\{n-1, n\}$ at the corner pair $\{c_X^-, c_X^+\}$. Both extensions have $\text{pos}(n) \in \{c_X^-, c_X^+\}$. Extending by linearity to V_{μ_X} , the seminormal support of $\Phi_X^\lambda(v)$ consists of such extensions, hence preserves the position-of- n support restriction. $\square$ $\square$

3.3 Cross-vanishings

Proposition 6 (Cross-projection vanishings at λ). *For all $i \in \{1, 2\}$ and $j \in \{1, 2, 3\}$,*

$$\pi_n^{(c_2)} F_A^i = 0, \quad \pi_n^{(c_1)} F_B^j = 0.$$

Proof. By Corollary 4, $\pi_n^{(c_2)} F_A^i = \pi_n^{(c_2)} \mathcal{O}_\lambda \Phi_A^\lambda(u_A^i) = \mathcal{O}_\lambda \pi_n^{(c_2)} \Phi_A^\lambda(u_A^i)$. By Lemma 5, every SYT in the seminormal support of $\Phi_A^\lambda(u_A^i)$ has $\text{pos}(n) \in \{c_1, c_3\}$, never c_2 . Hence $\pi_n^{(c_2)} \Phi_A^\lambda(u_A^i) = 0$, so $\pi_n^{(c_2)} F_A^i = 0$. Symmetrically for the F_B^j case. $\square$ $\square$

3.4 Conclusion: A and B are separated

Proof of Theorem 2. Suppose

$$\alpha_1 F_A^1 + \alpha_2 F_A^2 + \beta_1 F_B^1 + \beta_2 F_B^2 + \beta_3 F_B^3 = 0. \quad (2)$$

Apply $\pi_n^{(c_2)}$ to both sides. By Proposition 6, $\pi_n^{(c_2)} F_A^i = 0$ for both i . Hence

$$\beta_1 \pi_n^{(c_2)} F_B^1 + \beta_2 \pi_n^{(c_2)} F_B^2 + \beta_3 \pi_n^{(c_2)} F_B^3 = 0.$$

By hypothesis (b) of Theorem 2, the vectors $\{\pi_n^{(c_2)} F_B^j\}_{j=1}^3$ are linearly independent in $\pi_n^{(c_2)} V_\lambda$. Therefore $\beta_1 = \beta_2 = \beta_3 = 0$.

Substituting back into (2): $\alpha_1 F_A^1 + \alpha_2 F_A^2 = 0$. Apply $\pi_n^{(c_1)}$ to this equation: $\alpha_1 \pi_n^{(c_1)} F_A^1 + \alpha_2 \pi_n^{(c_1)} F_A^2 = 0$. By hypothesis (a), $\{\pi_n^{(c_1)} F_A^i\}_{i=1}^2$ are linearly independent. So $\alpha_1 = \alpha_2 = 0$.

Hence all five coefficients vanish, proving linear independence. Since these five vectors lie in $B_{j_0}^{(\lambda)} V_\lambda$ (by (1) and Definition 1), we have $\dim B_{j_0}^{(\lambda)} V_\lambda \geq 5$. $\square$ $\square$

Remark 7 (No c_3 -analysis needed). The proof does *not* require any analysis of the $\pi_n^{(c_3)}$ -components: the position-of- n projection alone suffices to decouple A from B , provided the within-class projection independences (a), (b) are established. The c_3 -direction, where both Φ_A^λ - and Φ_B^λ -supports overlap, plays no role.

Proposition 8 (Discarded sub-claim: NOT TRUE in general). *The naive sub-statement “ $\alpha_1 F_A^1 + \alpha_2 F_A^2 = 0$ and $\beta_1 F_B^1 + \beta_2 F_B^2 + \beta_3 F_B^3 = 0$ separately, from (2) alone” is stronger than Theorem 2 and is not directly provable from the position-of- n argument alone (both sides could lie non-trivially in the shared $\pi_n^{(c_3)} V_\lambda$). Theorem 2’s statement (via the hypothesis (a), (b)) is the correct, sufficient formulation.*

4 Inner-basis structure

4.1 The μ_A basis via the $r=1$ reduction

$\mu_A = (3, 3, 1^{q-1})$ has $\ell(\mu_A) = q+1$, $j_0(\mu_A) = q+2$, $|\mu_A| = q+5$. It has a unique length-preserving corner $c_1(\mu_A) = (2, 3)$, column-1 bottom $c_*(\mu_A) = (q+1, 1)$, and a same-row pair at row 2.

The $r=1$ reduction (May-13 r1 + SRD-331) reads

$$B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} = \mathcal{O}_{\mu_A} \Phi_{*}^{\mu_A} (B_{j_0(\mu_1^A)}^{(\mu_1^A)} V_{\mu_1^A}), \quad (3)$$

where $\mu_1^A := \mu_A \setminus \{c_1(\mu_A), c_*(\mu_A)\} = (3, 2, 1^{q-2})$ on $q+3$ letters, and $\mathcal{O}_{\mu_A} := (T_{q+1}+1)(T_{q+2}+1)(T_{q+3}+1)$ (three factors).

$\dim B_{j_0(\mu_1^A)}^{(\mu_1^A)} V_{\mu_1^A} = 2$ by the May-12 2026-05-12-dim-2-321.tex paper (structural at $|\mu_1^A| \geq 8$, i.e., $q \geq 5$; computational at $q \in \{3, 4\}$). Explicit basis

$$\tilde{F}_A^{\mu_1^A} := \mathcal{O}_{\mu_1^A} \Phi_A^{\mu_1^A} (u^{\nu_A^{\mu_1^A}}), \quad \tilde{F}_B^{\mu_1^A} := \mathcal{O}_{\mu_1^A} \Phi_B^{\mu_1^A} (u^{\nu_B^{\mu_1^A}}),$$

where $\nu_A^{\mu_1^A} = (2, 2, 1^{q-3})$ and $\nu_B^{\mu_1^A} = (3, 1^{q-2})$ on $q+1$ letters, $\mathcal{O}_{\mu_1^A} = (T_q+1)(T_{q+1}+1)$ (two factors), and $u^{(\nu)}$ denotes the $\dim-1$ generator of the j -formula image on ν .

Definition 9 (μ_A -basis).

$$u_A^1 := \mathcal{O}_{\mu_A} \Phi_{*}^{\mu_A}(\tilde{F}_A^{\mu_1^A}), \quad u_A^2 := \mathcal{O}_{\mu_A} \Phi_{*}^{\mu_A}(\tilde{F}_B^{\mu_1^A}).$$

By (3), $\{u_A^1, u_A^2\}$ spans $B^{(\mu_A)}$. The lower bound $\dim \geq 2$ requires showing $\mathcal{O}_{\mu_A} \Phi_{*}^{\mu_A}$ is injective on the 2-dimensional $B^{(\mu_1^A)}$; this is an open structural problem (verified computationally at $q \in \{3, 4, 5\}$, the SRD-331 paper’s range).

4.2 The μ_B basis from May-12-Eve3

$\mu_B = (4, 2, 1^{q-1})$ is the May-12 evening-3 family at $|\mu_B| = q + 5$ (with the paper’s $n = q + 5$). The three-vector basis

$$u_B^1 := F_{A,A}^{(\mu_B)}, \quad u_B^2 := F_{A,B}^{(\mu_B)}, \quad u_B^3 := F_B^{(\mu_B)}$$

is May-12-Eve3 Definition 5.1, with linear independence proved there (May-12-Eve3 §5.3, §6, §7; 2026-05-12-evening-4-2-1n-dim3.tex).

5 Why naive position-of-letter separators fail

5.1 The natural candidate: position-of- $(n - 2)$ within $\pi_n^{(c_1)}$

For the $(4, 3, 1^q)$ case, the within-A separation problem is: distinguish $\pi_n^{(c_1)} F_A^1$ from $\pi_n^{(c_1)} F_A^2$.

Both are seminormal vectors in $\pi_n^{(c_1)} V_\lambda$ (the subspace of V_λ supported on SYTs with letter $n = q + 6$ at $c_1 = (1, 4)$). Within this subspace, $n - 1 = q + 5$ has position $c_3 = (q + 2, 1)$ for all vectors in the support of $\pi_n^{(c_1)} \Phi_A^\lambda(\cdot)$ (the “B-direction” of Φ_A^λ).

The natural separator candidate: position-of- $(n - 2) = q + 4$.

Lemma 10 (Position-of- $(n - 2)$ is moved by both $(T_{n-3} + 1)$ and $(T_{n-2} + 1)$). T_j preserves position-of- $(n - 2)$ iff $j \notin \{n - 3, n - 2\}$.

Proof. T_j acts on the pair $\{j, j + 1\}$. Position-of- k is invariant under T_j iff $\{j, j + 1\} \cap \{k\} = \emptyset$, i.e., $j \neq k$ and $j \neq k - 1$. $\square$

Inside $\mathcal{O}_\lambda = (T_{q+2} + 1)(T_{q+3} + 1)(T_{q+4} + 1)(T_{q+5} + 1)$, the indices are $\{q + 2, q + 3, q + 4, q + 5\}$. Position-of- $(n - 2) = q + 4$ is moved by T_{q+3} (which acts on $\{q + 3, q + 4\}$) and by T_{q+4} (which acts on $\{q + 4, q + 5\}$). The factors $(T_{q+2} + 1)$ and $(T_{q+5} + 1)$ preserve it (the former touches $\{q + 2, q + 3\}$ only, the latter $\{q + 5, q + 6\}$).

5.2 Two-step contamination

The contamination pattern is structural and worth stating in general terms. Consider the recursive lift through three nested partitions:

$$\mu_1^A = (3, 2, 1^{q-2}) \subset \mu_A = (3, 3, 1^{q-1}) \subset \lambda = (4, 3, 1^q),$$

with intermediate sizes $q + 3, q + 5, q + 7$ respectively.

At each layer, the basis vectors $\tilde{F}_A^{(\mu_{\text{inner}})}$ and $\tilde{F}_B^{(\mu_{\text{inner}})}$ at the inner level are distinguished by the position of the inner top letter (call it k^{inner}) at some pair of exclusive cells.

Lemma 11 (Position-of-letter moves under outer chain). *For a chain $\mathcal{O} = (T_{k_1} + 1)(T_{k_2} + 1) \cdots (T_{k_r} + 1)$, the position of letter ℓ is preserved by T_{k_i} iff $\ell \notin \{k_i, k_i + 1\}$. Hence the cells reachable by the position-of- ℓ orbit after $\mathcal{O}$ form an “orbit” under the chain of moves $\{T_{k_i} : k_i = \ell - 1 \text{ or } \ell\}$.*

Proposition 12 (Contamination at the λ level). *For the natural inner separator at every layer, the outer chain $\mathcal{O}_\lambda$ at the λ level enlarges the position-of-letter orbit to overlap with the orbit of the other inner basis vector. Consequently, no single position-of-letter projection $\pi_k^{(c)}$ at the λ level is exclusive between F_A^1 and F_A^2 (or between any F_X^i pair sharing the same Φ_X^λ class).*

Proof sketch. The position of any specific letter k at the λ level is determined by:

1. Its position in the original μ_1^A - (or μ_B -)level basis vector.
2. Possible moves through $\mathcal{O}_{\mu_1^A}$ at the innermost layer (two $(T_j + 1)$ factors).
3. Lift through $\Phi_*^{\mu_A}$ (or analog).
4. Moves through $\mathcal{O}_{\mu_A}$ at the middle layer (three factors).
5. Lift through Φ_X^λ .
6. Moves through $\mathcal{O}_\lambda$ at the outer layer (four factors).

Total: $2 + 3 + 4 = 9$ $(T_j + 1)$ factors in the cumulative chain. Each T_j -move can replace position-of-letter- k with the current position of letter $k - 1$ or $k + 1$ (depending on the index match with the move).

Starting from any exclusive cell at the innermost layer, the 9-step orbit reaches a set of cells that, for typical Hoefsmit non-degenerate configurations, includes the originally-exclusive cell of the OTHER inner basis vector. Concretely: the cells $(1, 3)$ and $(2, 2)$ are at axial distance $c(1, 3) - c(2, 2) = 2$, which yields a non-degenerate T_j -2-block for appropriate j , and this move *is* reachable somewhere in the 9-step chain. Hence the contamination occurs generically.

A fully rigorous proof would enumerate the 9-step orbit explicitly (a finite case analysis, parallel to May-12-Eve3 §5.4 but expanded by an order of magnitude due to the additional recursion depth). We omit the enumeration; the structural conclusion suffices for the discussion. $\square$ $\square$

Remark 13 (Cost of recursion depth). The recursion depth of the contamination is structurally tied to the shape:

- Hooks (e.g. $(k, 1^*)$): one layer of recursion; clean position-of- $(n - 1)$ separators (May-12 hook paper).
- $r = 2$ shapes without long inner non-hook chains (e.g. $(4, 2, 1^*)$): two layers; position-of- $(n - 1)$ + position-of- $(n - 2)$ suffice (May-12-Eve3).
- $r = 2$ same-row shapes like $(4, 3, 1^*)$: three layers (the μ_A -side has a same-row pair triggering an extra $r=1$ lift). The 9-step chain contaminates the natural separator.

This suggests a coarse rule: *position-of-letter techniques exhaust at depth ≤ 2* . Deeper recursions require finer tools.

6 Path forward: tracer-pair technique

The memory note `tracer-pair-technique` (this session) records a successful method from the hook program:

Tracer-pair: when $\dim \leq d$ is known, track d matrix coefficients between explicit SYTs; fixed entries force unique contributors.

6.1 Applied to $(4, 3, 1^q)$ lower bound

Choose explicit λ -SYTs $T_1^*, \dots, T_5^*$ and compute the 5×5 “tracer matrix”

$$M_{ik} := \langle v_{T_i^*}, F_X^{j_k} \rangle$$

where (X, j_k) runs through the 5 basis indices $\{(A, 1), (A, 2), (B, 1), (B, 2), (B, 3)\}$. If $\det M \in \mathbb{Q}(q)^\times$, then F ’s are linearly independent.

6.2 Reduction via the A–B decoupling (Theorem 2)

By Theorem 2 (the A vs B separation), the 5×5 tracer matrix has block-diagonal structure: choose two T_i^* ’s with $\text{pos}(n) = c_1$ (capturing only F_A ’s) and three with $\text{pos}(n) = c_2$ (capturing only F_B ’s). Then M splits into a 2×2 block (for F_A ’s) and a 3×3 block (for F_B ’s).

6.3 The two reduced problems

2×2 tracer for F_A^1, F_A^2 . Pick two λ -SYTs T_1^*, T_2^* with $\text{pos}(n) = c_1$. Distinguish them by some interior letter pattern that detects the inner $\tilde{F}_A^{\mu_1^A}$ vs $\tilde{F}_B^{\mu_1^A}$ distinction. Concretely: T_1^* has interior column-2 entry $\gamma \in \{q, q+1, q+2\}$ (from the u^{ν_A} support, before all the chains); T_2^* has interior row-1 entry $b = q+2$ (distinguishing the $\tilde{F}_B^{\mu_1^A}$ -derived inner basis).

3×3 tracer for F_B^1, F_B^2, F_B^3 . Pick three λ -SYTs T_3^*, T_4^*, T_5^* with $\text{pos}(n) = c_2$. The May-12-Eve3 separation (column-2 entries of the inner $\mu_A^{\mu_B} = (3, 2, 1^{q-2})$ shape vs row-1 entries of the inner $\mu_B^{\mu_B} = (4, 1^{q-1})$ shape) provides three clean SYT supports.

6.4 Status

The tracer-pair technique reduces the structural lower bound to *five explicit Hoefsmit coefficient computations* (the $5 \times 5 = 4 + 9$ entries of the 2×2 and 3×3 blocks). Each entry is a product of:

1. One “inner-inner” Hoefsmit coefficient (from $\mathcal{O}_{\mu_1^A}$ acting on u^ν).
2. One “inner” Hoefsmit coefficient (from $\mathcal{O}_{\mu_A}$ or $\mathcal{O}_{\mu_B}$).
3. One “outer” Hoefsmit coefficient (from $\mathcal{O}_\lambda$).
4. Several τ, τ_X factors from the Φ -maps.

The non-vanishing of the 2×2 and 3×3 determinants reduces to algebraic non-vanishing of explicit polynomial expressions in q . This is a finite computation, verifiable at any specific q and (with care) provable structurally via the $b' = 1$ propagation technique (memory note **b-prime-propagation**).

7 Computational verification

From the May-13-UB paper Section 5 (verbatim):

λ	$\dim V_\lambda$	$\dim B_{j_0}^{(\lambda)} V_\lambda$
$(4, 3, 1, 1, 1) \quad (q = 3, n = 10)$	525	5
$(4, 3, 1, 1, 1, 1) \quad (q = 4, n = 11)$	1100	5

At both verified cases, $\dim = 5$, consistent with the conjectural lower bound from the five-vector construction.

A separate, more refined computational check (not yet performed in this session): compute the explicit five vectors $F_A^1, F_A^2, F_B^1, F_B^2, F_B^3$ via the chain of operations defined in Section 2 and verify that the resulting 5-vector matrix has rank 5. This would also exhibit the supports and enable a precise tracer-pair selection.

8 Honest status

What we proved structurally.

1. The explicit five-vector construction (Definition 1).
2. Position-of- n separation between A-pieces and B-pieces (Proposition 6).
3. The full decoupling statement (Theorem 2 + proof in §3): the 5-vector independence reduces to two within-class projection independences, without any analysis of the $\pi_n^{(c_3)}$ -component (Remark 7).

What we identified.

1. The naive position-of-letter separator (analogous to May-12-Eve3 Lemma 5.4) does *not* propagate cleanly through multi-layer recursions (Proposition 11). The $(4, 3, 1^q)$ shape requires three layers of recursion, and the chain of T_j moves contaminates the natural separator at every layer.
2. The correct technique is *tracer-pair* via explicit Hoefsmit coefficient computation (Section 6). The reduction splits the 5-vector problem into a 2×2 and a 3×3 tracer computation.

What remains.

1. Execute the 2×2 tracer for $\{F_A^1, F_A^2\}$ — five nested Hoefsmit factors, structural at $q \geq q_*$, computational at small q . Reduces to the structural lower bound for $\mu_A = (3, 3, 1^{q-1})$, currently open (SRD-331 Remark 3.4).
2. Execute the 3×3 tracer for $\{F_B^1, F_B^2, F_B^3\}$ — directly lifts the May-12-Eve3 position-of- $(n-1)$ analysis, plus the outer factor $(T_{n-5} + 1)$ specific to $(4, 3, 1^q)$.

(The c_3 -direction analysis is *not* needed — Remark 7 — only the two within-class projection independences above.)

Combined with May-13-UB. If the three remaining items are closed, we obtain $\dim B_{j_0}^{(\lambda)} V_\lambda = 5$ structurally for all $q \geq q_*$ (with q_* to be determined by the chain of structural ranges).

9 Comparison with May-12-Eve3

The May-12-Eve3 paper proved $\dim B^{((4,2,1^{n-6}))} = 3$ by the analogous strategy:

- Two-vector A-piece $F_{A,A}, F_{A,B}$ via inner basis from May-12-today on $\mu_A = (3, 2, 1^{m-5})$.
- One-vector B-piece F_B via inner basis from May-12 hook on $\mu_B = (4, 1^{m-4})$.
- A vs B separation via position-of- n .

- Within-A separation via position-of- $(n-1)$, with a single T_{n-2} outer-factor move analysed via Lemma posnm1.

The key technical step that worked there: the outer chain had *three* factors $(T_{n-4} + 1)(T_{n-3} + 1)(T_{n-2} + 1)$, and the position-of- $(n-1)$ was preserved by the first two factors. The single T_{n-2} -move was sufficient to track and the reachable cells were a 2-element set with clean exclusive elements.

For $(4, 3, 1^q)$, the outer chain has *four* factors. The analogous separator position-of- $(n-2)$ would be preserved by only two of the four factors, and the two outer-factor moves enlarge the reachable set beyond the 2-element “clean” structure.

This is a structural difficulty inherent to the shape, not a defect of the proof method. It explains why the May-12-Eve3 technique does not directly translate, and motivates the tracer-pair refinement.